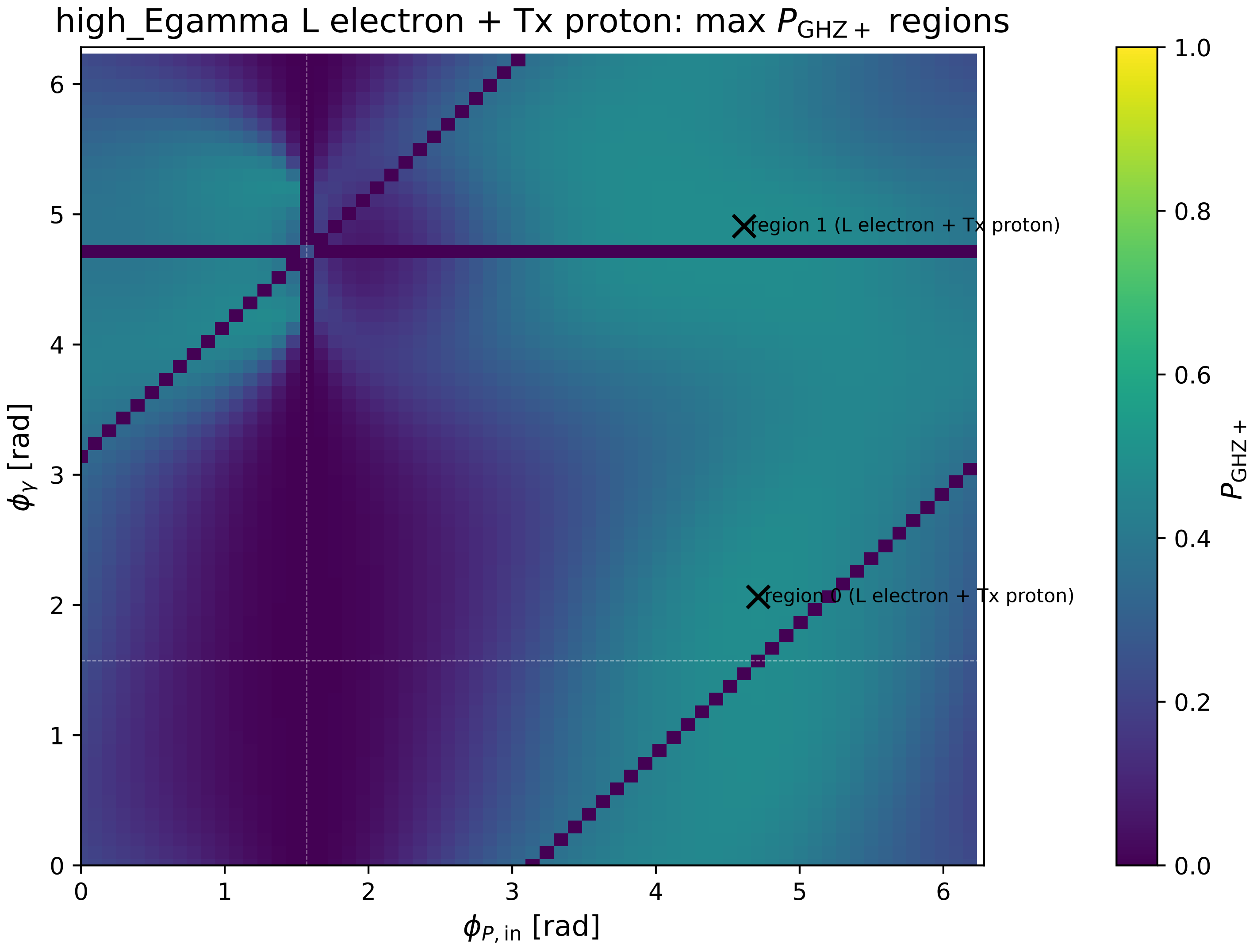
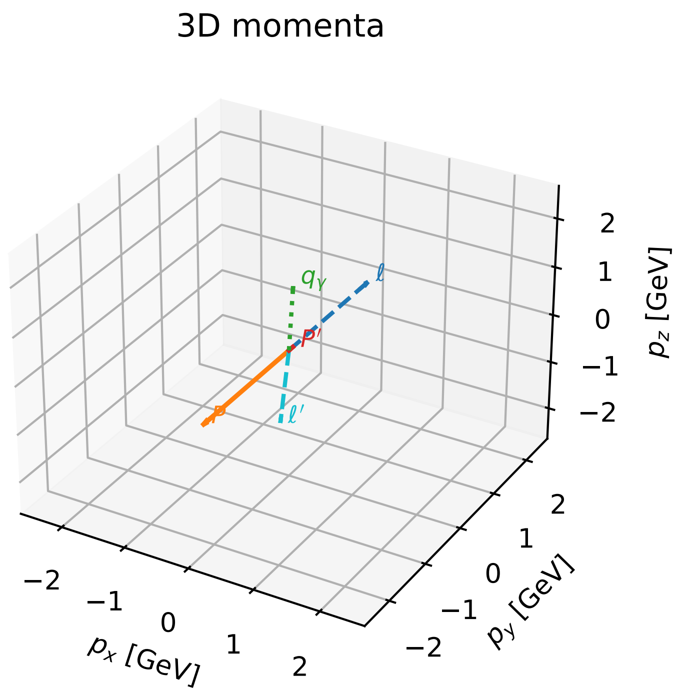

high_Egamma L electron + Tx proton: max $P_{\mathrm{GHZ}+}$ regions
region 1 (L electron + Tx proton)
region 0 (L electron + Tx proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$P_{\mathrm{GHZ}+}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

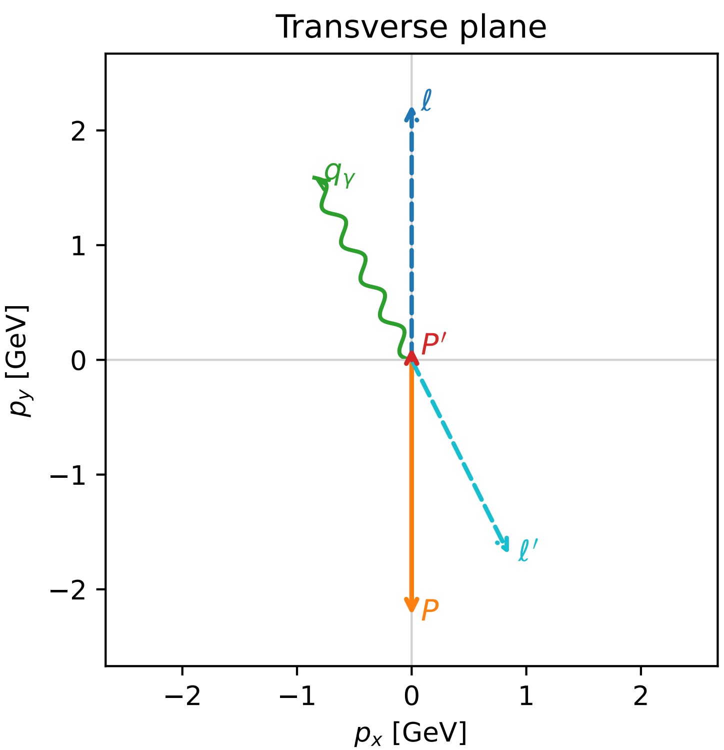
L electron + Tx proton characteristic max $P_{\rm GHZ+}$ configuration

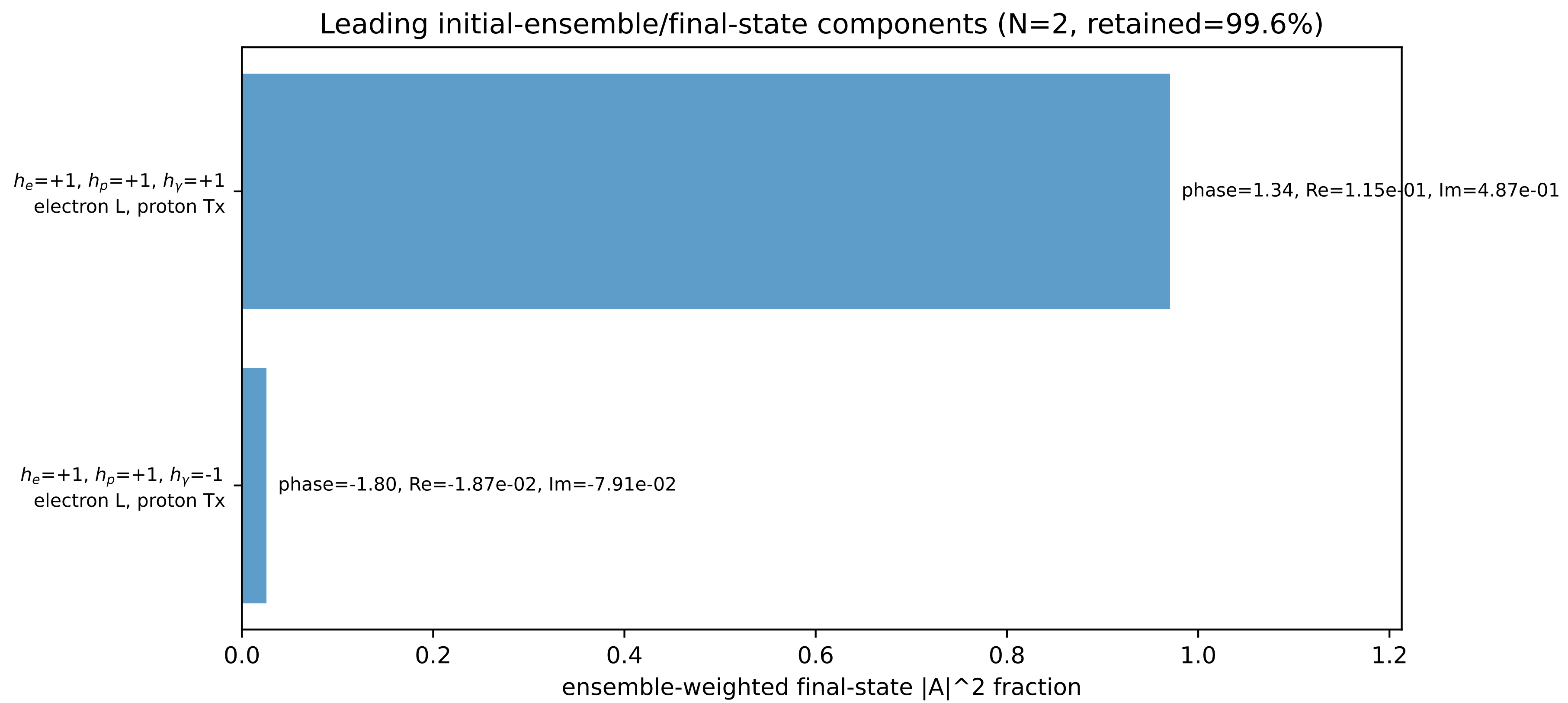


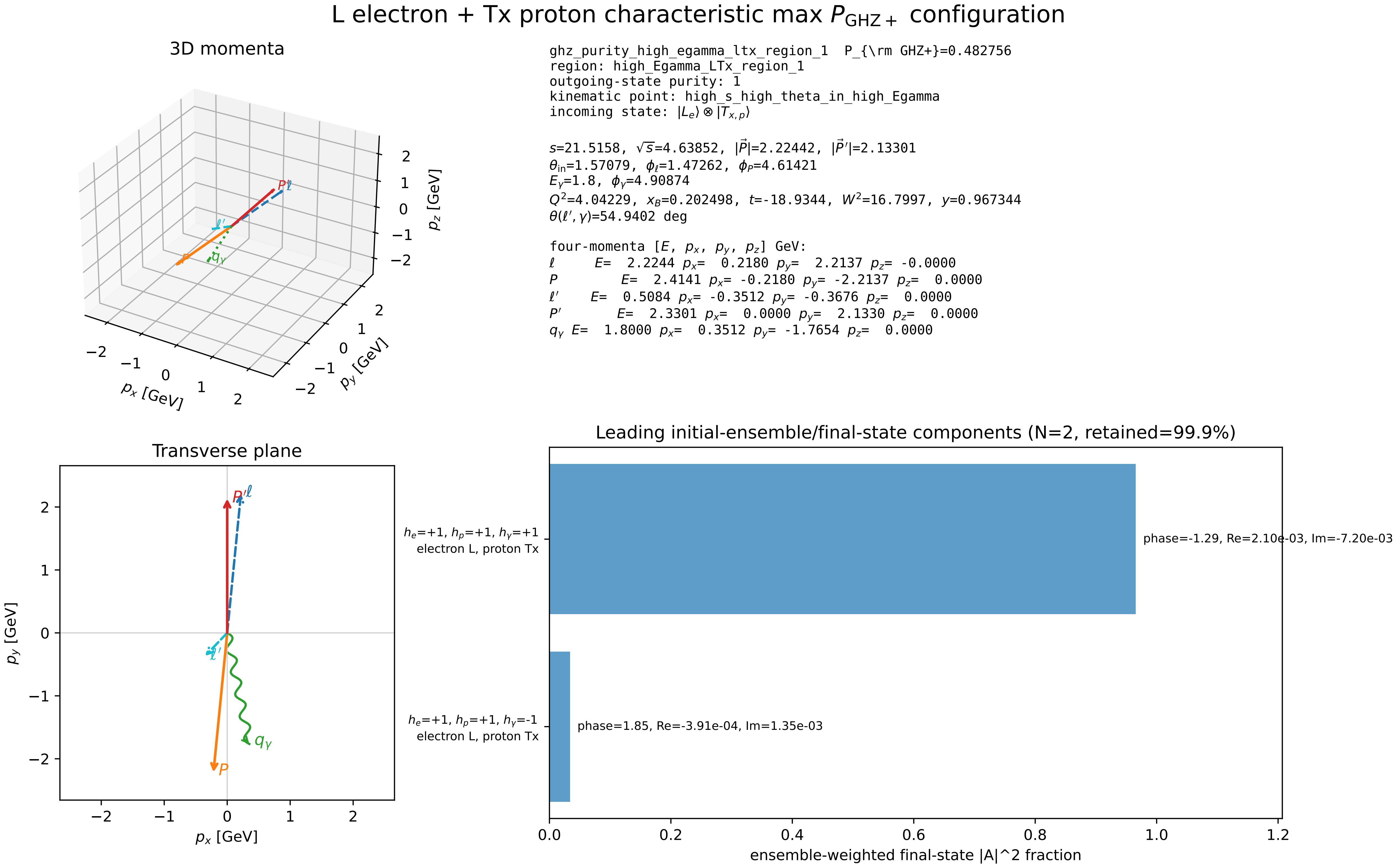
ghz_purity_high_egamma_ltx_region_0 P_{\rm GHZ+}=0.485167
region: high_Egamma_LTx_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_e\rangle \otimes |T_{x,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.106402
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1.8, ϕ_γ=2.06167
Q^2=15.964, x_B=0.839123, t=-3.27157, W^2=3.94048, y=0.921917
$\theta(\ell',\gamma)$=178.483 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 1.8945 p_x= 0.8485 p_y= -1.6939 p_z= 0.0000
P' E= 0.9440 p_x= 0.0000 p_y= 0.1064 p_z= 0.0000
q_γ E= 1.8000 p_x= -0.8485 p_y= 1.5875 p_z= 0.0000






L electron + Tx proton characteristic max $P_{\rm GHZ+}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
ghz_purity_high_egamma_ltx_region_1 P_{\rm GHZ+}=0.482756
region: high_Egamma_LTx_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_e\rangle \otimes |T_{x,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.13301
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.47262, ϕ_P=4.61421
E_γ=1.8, ϕ_γ=4.90874
Q^2=4.04229, x_B=0.202498, t=-18.9344, W^2=16.7997, y=0.967344
$\theta(\ell', \gamma)$=54.9402 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.2180 p_y= 2.2137 p_z= -0.0000
P E= 2.4141 p_x= -0.2180 p_y= -2.2137 p_z= 0.0000
ℓ' E= 0.5084 p_x= -0.3512 p_y= -0.3676 p_z= 0.0000
P' E= 2.3301 p_x= 0.0000 p_y= 2.1330 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.3512 p_y= -1.7654 p_z= 0.0000
Transverse plane
Leading initial-ensemble/final-state components (N=2, retained=99.9%)
h_e=+1, h_p=+1, h_γ=+1
electron L, proton Tx
phase=-1.29, Re=2.10e-03, Im=-7.20e-03
h_e=+1, h_p=+1, h_γ=-1
electron L, proton Tx
phase=1.85, Re=-3.91e-04, Im=1.35e-03
ensemble-weighted final-state |A|^2 fraction